

Erdős #700: the capped carry ledger for $\min_k \gcd(n, \binom{n}{k})$

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Abstract

For $f(n) = \min_{1 < k \leq n/2} \gcd(n, \binom{n}{k})$ we give the exact form of the minimand as a product of per-prime carry counts *capped* at $v_p(n)$, deduce $f(n) \mid n$ and $f(n) \leq \text{rad}(n)$, and explain from the cap why the measured set $\{n : f(n) > \sqrt{n}\}$ is squarefree-dominated. #700 is not proved.

1 The capped ledger

Let $\kappa_p(k, n-k)$ denote the number of carries in the base- p addition $k + (n-k)$, so that $v_p(\binom{n}{k}) = \kappa_p$ by Kummer.

Proposition 1. *For n composite and $1 < k \leq n/2$,*

$$\gcd\left(n, \binom{n}{k}\right) = \prod_{p^a \parallel n} p^{\min(a, \kappa_p(k, n-k))}, \quad f(n) = \min_{1 < k \leq n/2} \prod_{p^a \parallel n} p^{\min(a, \kappa_p)}.$$

In particular $f(n) \mid n$, and $f(n) \leq \text{rad}(n)$ when n is squarefree.

Proof. The valuation of a gcd is the minimum of the valuations, and $v_p(n) = a$, $v_p(\binom{n}{k}) = \kappa_p$. Each exponent is at most a , so the product divides n . If n is squarefree every $a = 1$ and each factor is p^0 or p^1 . $\square$

Verified: $f(n) \mid n$ with no exception for all composite $n < 300$.

Remark 2 (why the cap matters). *This is the one member of the $\binom{n}{k}$ domination cluster whose rails saturate: rail p contributes at most $v_p(n)$ however many carries occur. Elsewhere in the cluster (#683, #685, #1094, #1095) the relevant statistic is unbounded in the carry count, and the analysis is driven by which rails carry at all; here it is driven by which rails can be made not to carry, simultaneously, by a single choice of k .*

Corollary 3. *For composite n , taking $k = p^a$ with $p^a \parallel n$ gives $\gcd(n, \binom{n}{p^a}) = n/p^a$, whence $f(n) \leq n/P(n)$ where $P(n)$ is the largest prime factor, and therefore $f(n) \leq (1+o(1))n/\log n$.*

This is the bound recorded by Erdős and Szekeres; it is included because Proposition 1 makes it a one-line valuation computation.

2 Prime powers, and the correct divisor

Erdős and Szekeres record $f(n) \leq n/P(n)$ from the choice $k = p^a$ with $p^a \parallel n$. That choice is legitimate only when $p^a \leq n/2$, and tracking the restriction determines the prime-power case completely.

Theorem 4. *Let p be prime and $a \geq 2$. Then $f(p^a) = p$.*

Proof. Put $n = p^a$ and let $1 < k \leq n/2$. Since $s_p(n) = 1$, Legendre and Kummer give

$$v_p\binom{n}{k} = \kappa_p = \frac{s_p(k) + s_p(n-k) - 1}{p-1},$$

and n has the single nonzero digit 1 in position a , so any addition $k + (n-k) = p^a$ with $0 < k < p^a$ carries at least once; hence $\kappa_p \geq 1$ and $p \mid \binom{n}{k}$ for every such k . As p is the only prime dividing n , this gives $\gcd(n, \binom{n}{k}) = p^{\min(a, \kappa_p)} \geq p$, so $f(p^a) \geq p$.

For the reverse, take $k = p^{a-1}$, which satisfies $1 < k \leq n/2$ because $a \geq 2$ and $p \geq 2$. Then $s_p(k) = 1$ and $n-k = p^{a-1}(p-1)$ has the single digit $p-1$, so $s_p(n-k) = p-1$ and

$$\kappa_p = \frac{1 + (p-1) - 1}{p-1} = 1.$$

Hence $\gcd(n, \binom{n}{p^{a-1}}) = p^{\min(a, 1)} = p$, so $f(p^a) \leq p$. □

Verified: all prime powers $p^a \leq 2500$ with $p \leq 13$, $a \geq 2$.

Proposition 5. *Let n be composite and let $q = p^{v_p(n)}$ be any prime power exactly dividing n with $q \leq n/2$. Then $f(n) \leq n/q$. Consequently*

$$f(n) \leq \frac{n}{Q'(n)}, \quad Q'(n) := \max\{p^{v_p(n)} : p^{v_p(n)} \leq n/2\}.$$

Proof. Take $k = q$ in the minimum defining $f(n)$; the constraint $1 < k \leq n/2$ holds by hypothesis. Writing $q = p^a$ with $a = v_p(n)$, Kummer gives $\kappa_p(q, n-q) = a$, since adding q to $n-q$ in base p carries out of each of the positions $0, \dots, a-1$ and no further, while for $\ell \neq p$ one has $\kappa_\ell \geq 0$. Hence $\gcd(n, \binom{n}{q}) = n/q$, and $f(n)$, being a minimum, is at most this. □

Verified: 1095 pairs (n, q) with $n < 600$.

Corollary 6. *If n is composite, is not the square of a prime, and satisfies $f(n) = n/P(n)$, then $P(n) = Q'(n)$: the largest prime factor of n is also its largest admissible prime power.*

Proof. By Proposition 5, $f(n) \leq n/Q'(n) \leq n/P(n)$ whenever $P(n) \leq n/2$, which holds for composite n other than $n = p^2$. Equality $f(n) = n/P(n)$ therefore forces $Q'(n) \leq P(n)$, and $Q'(n) \geq P(n)$ always, since $P(n)$ itself is an admissible prime power in that range. □

Verified: the exceptions to $P = Q'$ among $n < 600$ with $f(n) = n/P(n)$ are exactly the prime squares 4, 9, 25, 49, 121, 169, 289, 361, 529, and no others.

3 Measurement

Census over composite $n < 600$:

- $f(n) = n/P(n)$ for 307 values, beginning 4, 6, 9, 10, 14, 15, 20, 21, 22, 25, 26, 28, 30, 33 — consistent with the known cases (products of two primes, and $n = 30$).
- $f(n) > \sqrt{n}$ for 30 values: 30, 70, 84, 105, 132, 140, 154, 165, 168, 182, ... with $f = 6, 10, 12, 15, 12, 14, 14, 15, 9, 10, 12, 14, 15, 16, 18, 20, 21, 24, 25, 28, 30$.
The count is still growing at the top of the range.

The second list is squarefree-dominated, which Proposition 1 explains: for $f(n)$ to be large every rail must reach its cap for *every* k , and a squarefree n with several small prime factors is the configuration in which all caps are 1 and can be attained together.

Remark 7 (register). ***Proved:** Proposition 1 and Corollary 3, elementary from Legendre and Kummer; neither is claimed new (Corollary 3 is Erdős–Szekeres). Theorem 4 determines f on prime powers, $f(p^a) = p$ for $a \geq 2$; Proposition 5 replaces $P(n)$ by the largest admissible prime power $Q'(n)$; and Corollary 6 gives a necessary condition, $P(n) = Q'(n)$, for the equality $f(n) = n/P(n)$ that Erdős asks to characterise, with the prime squares as the exact exceptions. **Measured:** the two censuses. **Not proved:** the characterisation of $\{n : f(n) = n/P(n)\}$, the infinitude of $\{n : f(n) > \sqrt{n}\}$, or the bound $f(n) \ll_A n/(\log n)^A$. **Falsifier:** $\{n : f(n) > \sqrt{n}\}$ terminating at larger n would contradict the apparent growth. **Next:** census to $n = 10^5$ and test whether that set has positive density; and characterise $\{f(n) = n/P(n)\}$ by $\Omega(n)$ and squarefreeness, where the data suggests a clean statement is available.*